



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Lecture 2: DeFi 1

Tokenization and Stablecoin Design

SC6125 – Blockchain in Finance

Token Types (By Issuance Method)

Token Type	Issuance Authority	Backing / Basis	Key Real-Life Examples
Native Protocol Tokens	Blockchain protocol	None	Bitcoin (BTC), Ether (ETH)
Entity-Issued Unbacked Tokens	Company / DAO	None	UNI, governance tokens
Asset-Backed RWA Tokens	SPV / fund / issuer	Legal claim on off-chain assets	Securities, Tokenised bonds (Project Guardian), BlackRock BUIDL
Balance-Sheet Tokens	Regulated FI	Issuer liability	Tokenised bank deposits
Reserve-Backed Tokens (Stablecoins)	Stablecoin issuer	Segregated reserves	USDC, USDT, PAXG

Token Types (By Issuance Method)

Token Type	Issuance Authority	Backing / Basis	Key Real-Life Examples
Wrapped / Custodial Tokens	Custodian / bridge	Locked underlying token	WBTC, wETH
Algorithmic Tokens (Crypto-Collateralised)	Protocol logic	Over-collateralised crypto assets	DAI, crypto-backed synthetics
Algorithmic Tokens (Non-Collateralised)	Protocol logic	Incentives & reflexivity	Rebasing tokens, seigniorage designs
Synthetic Tokens	Protocol + collateral + oracles	Collateralised economic exposure	Synthetic USD, synthetic equities
Credential / Non-Transferable Tokens	Trusted institution	Attestation	On-chain KYC or accreditation tokens

How to tokenize real world assets (incl securities)?

- There is no standardize process (yet)
- There are 8 commonly used steps:
 1. Asset & Rights Structuring (Legal First, Not Tech)
 - Objective: Define what is actually being tokenized
 2. Regulatory Classification & Jurisdictional Analysis
 - Objective: Determine applicable laws and licensing
 3. Legal Vehicle Formation
 - Objective: Create a legally enforceable wrapper
 4. Offering Documentation & Disclosure
 - Objective: Make it investable and compliant.



How to tokenize real world assets (incl securities)?

- There are 8 commonly used steps:
 5. Token Design & Smart Contract Architecture
 - Objective: Translate legal rights into programmable logic
 6. Issuance, Distribution & Onboarding
 - Objective: Issue tokens to compliant investors
 7. Post Issuance Lifecycle Management
 - Objective: Operate the asset over time
 8. Secondary Trading, Settlement & Redemption
 - Objective: Enable liquidity while remaining compliant

Legal → Regulatory → Structural → Technical → Operational



RWA Tokenization (Example: Gold-Backed Tokens)

Use Case & Problem Definition

- The core problem is operational trust fragmentation in existing gold markets across vault operators, custodians, brokers, clearing agents, and registrars, each maintaining separate ledgers
- RWA tokens enables digital representation, transfer, and redemption of vaulted physical gold with:
 - Verifiable backing (ounces in a vault $\leftrightarrow$ tokens in circulation)
 - Atomic issuance and redemption
 - Programmable transfer constraints
 - Near-real-time settlement

RWA Tokenization (Example: Gold-Backed Tokens)

Why blockchain is appropriate

- Blockchain directly addresses ledger integrity and coordination, not gold storage itself. It provides:
 - A single source of truth for token supply
 - Immutable issuance and redemption records
 - Atomic settlement (DvP-like behavior)
 - Verifiable constraints (no mint without backing)

RWA Tokenization (Example: Gold-Backed Tokens)

On-chain components

- Token contract (GoldToken)
 - Track balances (e.g., 1 token = 1 gram)
 - Enforce transfer restrictions
 - Support mint and burn
- Issuance & Redemption Controller
 - Validate mint requests
 - Enforce 1:1 backing invariant
 - Emit audit-relevant events
 - Trust assumption:
 - Controller keys correspond to legally accountable entity
- Governance / Emergency Control Contract
 - Pause transfers
 - Pause mint/burn
 - Upgrade contracts (if proxy-based)
 - Trust assumption:
 - Multisig or regulated entity governance

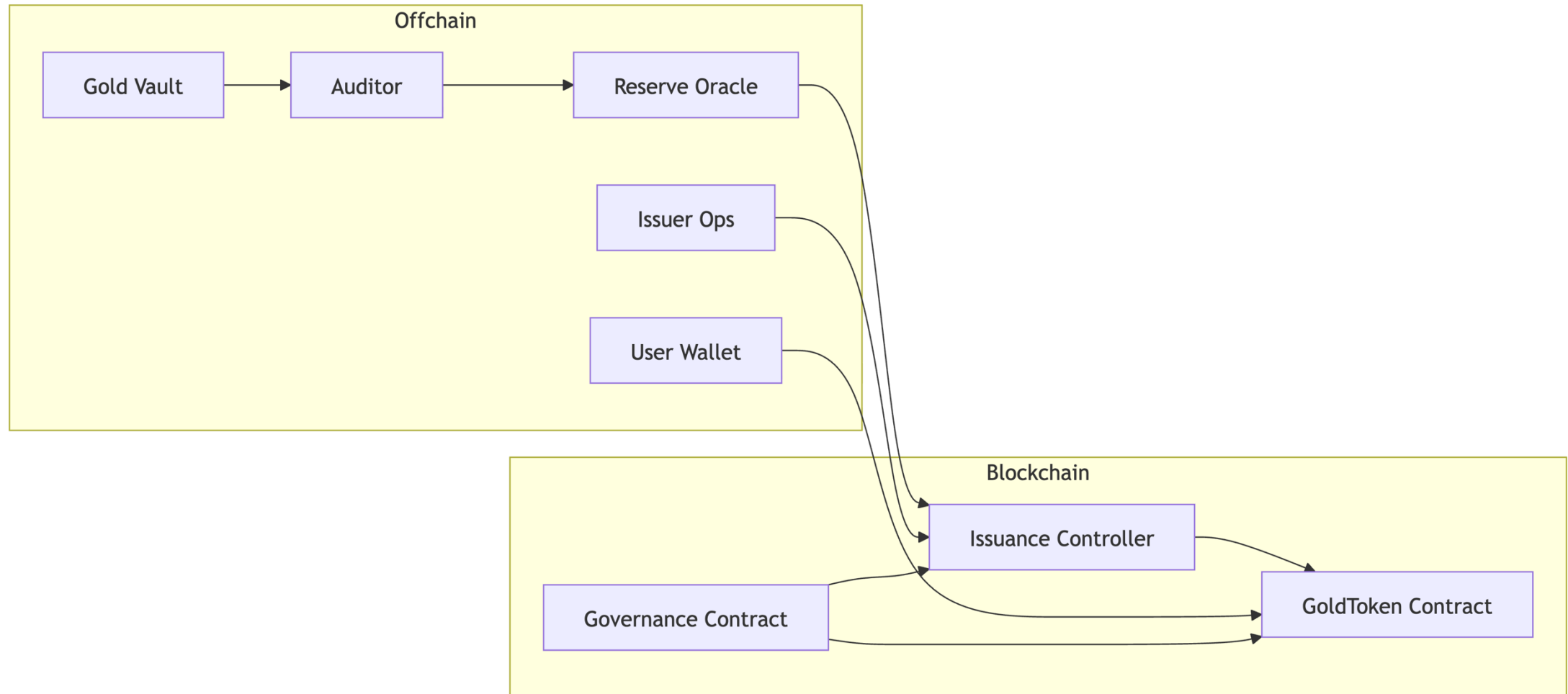
RWA Tokenization (Example: Gold-Backed Tokens)

Off-chain components

- Gold Vault management system
 - Tracks bar serials & weight
- Auditor
 - Confirms vault totals
- Reserve Oracle
 - Attests gold balances
- Issuer Ops
 - Coordinates legal & operational steps
- User wallet
 - Token interaction

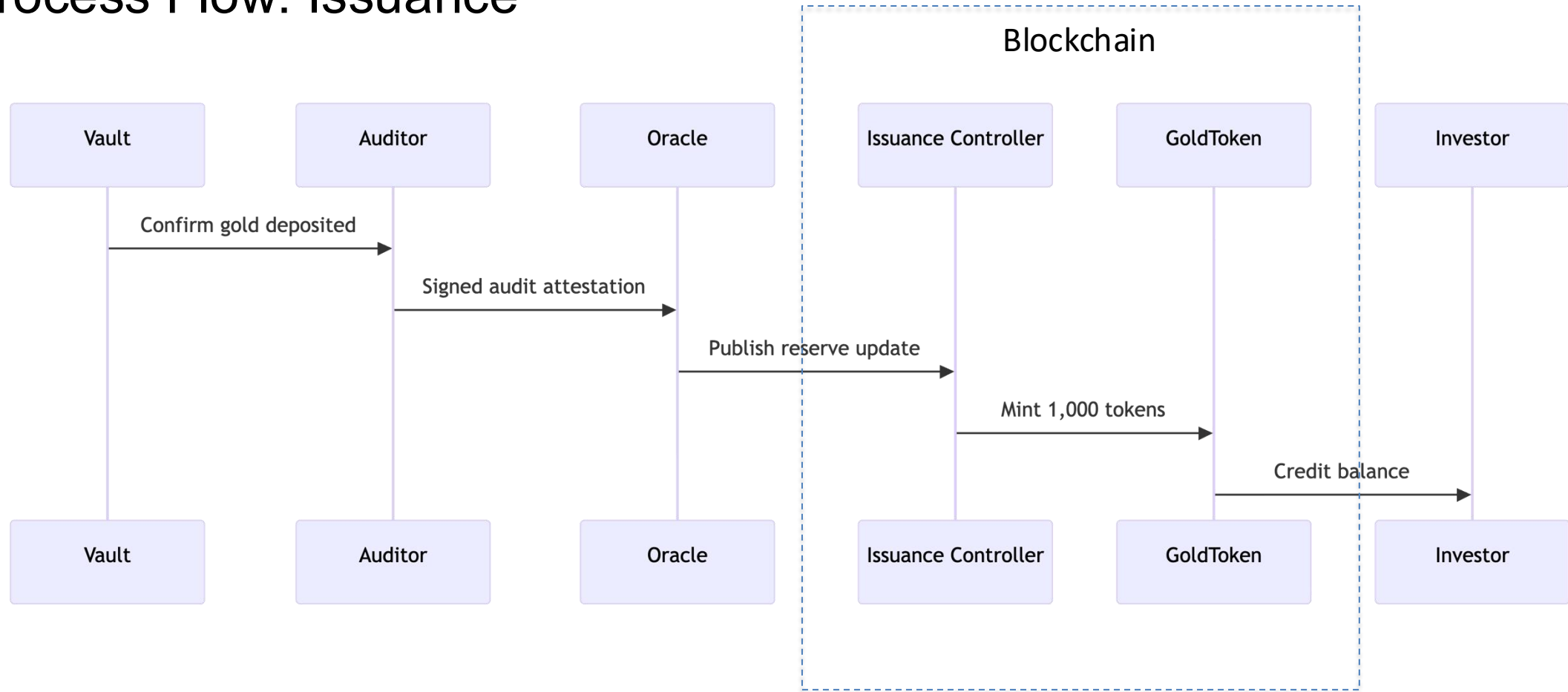
RWA Tokenization (Example: Gold-Backed Tokens)

Architecture



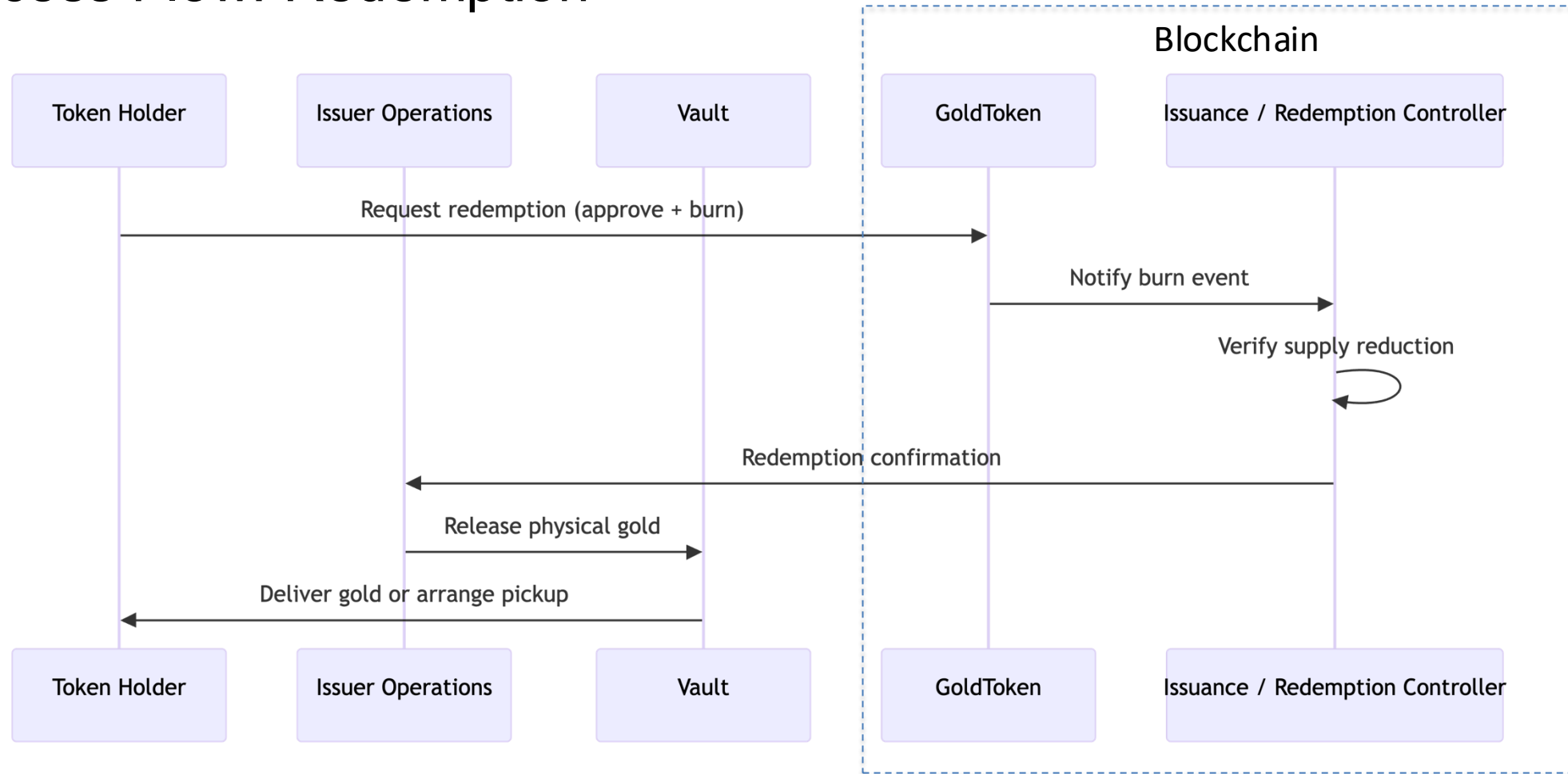
RWA Tokenization (Example: Gold-Backed Tokens)

Process Flow: Issuance



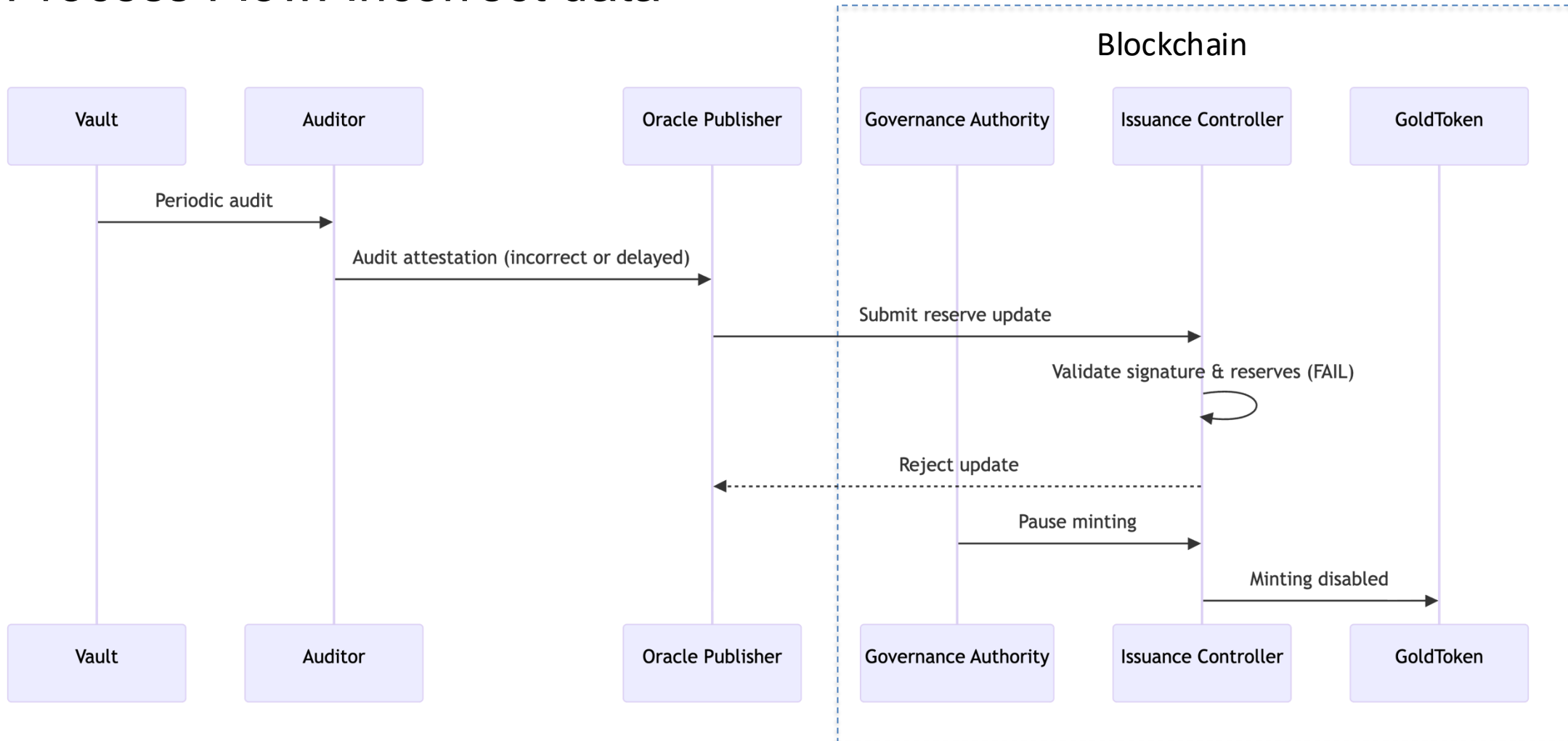
RWA Tokenization (Example: Gold-Backed Tokens)

Process Flow: Redemption



RWA Tokenization (Example: Gold-Backed Tokens)

Process Flow: Incorrect data



RWA Tokenization (Example: Gold-Backed Tokens)

- Incentives
 - Issuer: Reputation, legal liability
 - Auditor: Professional liability
 - Vault: Custody fees
 - Token holders: Redeemability
- Governance
 - Contract upgrades
 - Parameter changes
 - Emergency pause
 - Wind-down initiation
 - Using:
 - Multisig
 - Regulated entity board
 - Jurisdictional oversight

RWA Tokenization (Example: Gold-Backed Tokens)

- Benefits
 - Continuous, verifiable supply integrity
 - Atomic issuance and redemption
 - Programmable restrictions (jurisdiction, sanctions)
 - Reduced reconciliation risk
 - Instant settlement of ownership changes
- Trade-offs
 - Complexity: Higher operational overhead
 - Cost: Blockchain + audit + oracle
 - Governance: Requires strong control processes
 - UXWallet: management complexity

Fractionalizing Investment Securities

- FinTech firms and even conventional brokerages are beginning to sell fractionalized securities to their clients
- FinTech firms Robinhood and SoFi, as well as brokerages Fidelity and Charles Schwab, are now selling fractionalized equity securities for as little as \$1–\$5
- Most FinTech firms and conventional brokerages are not taking advantage of blockchain technologies
 - Companies buy whole securities and selling undivided fractional interests therein to their customers
 - The companies themselves typically “maintain a separate account” to record each customer’s purchased fractional interests
 - There are no exchanges on which to trade those fractional interests

Schwarcz, S. L., & Bourret, R. (2023). Fractionalizing investment securities: Using FinTech to expand financial inclusion. *Ohio State Law Journal*, 84.



Fractionalizing Investment Securities

- Tokenized securities “can be programmed with unique characteristics and ownership rights,” allowing investors to acquire “fractional ownership of the tokenized security’s underlying asset” (such as a bond or stock)
 - Smart contracts running on DeFi protocols can facilitate low-cost and efficient trading of tokenized fractionalized securities
- For example, FinTech company BondbloX is beginning to buy corporate bonds and sell fractionalized interests in the form of blockchain-recorded tokens, to investors
 - This new method of fractionalizing securities promises to have numerous advantages, including lowering costs for issuers and increasing transparency for investors

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Fractionalizing Investment Securities

Benefits for Investors

- Ability to invest in stocks and bonds notwithstanding the typical minimum investment amount, which ordinarily would be outside many retail investors' purchasing capacity
 - For example, the median account balance for a Robinhood retail investor is only \$240
 - BondBloX explains that although the minimum investment size for a bond of HSBC Bank is \$200,000, it has been able to sell \$1,000 fractionalized interests in bonds to a wide range of investors
- Investors can also diversify their investment portfolios, thereby reducing their risk

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Fractionalizing Investment Securities

- Singapore-based bank DBS was one of the first corporations to engage in a bond-based tokenized fractionalized security offering
 - A DBS Digital Bond was issued for \$11.3 million with a 6-month expiry paying 0.6% interest per year, and each fractionalized bond was traded in increments of \$7,600
 - The fractionalized bonds were issued in compliance with relevant jurisdictional laws, meaning holders of the bonds have the “same legal certainties and protections over their rights as [holders of] traditional bonds.”
- Societe Generale SFH, a French bank, also experimented with issuing bonds via tokenized fractionalized securities
 - While not a public offering, the fractionalized securities represented “a covered bond, a security that is backed by a pool of assets on the issuer’s balance sheet,” with a 5-year maturity subject to a 1-year extension
 - Moody’s and Fitch, two leading credit rating agencies, gave the securities the highest-possible credit rating of AAA
 - As key factors, Moody’s noted that the issuance of bonds in this manner lowered the likelihood of errors due to a reduced number of intermediaries and increased transparency through the use of blockchain technology

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Fractionalizing Investment Securities

- Risks and considerations
 - Liquidity risks
 - Credit risks
 - Intermediary risks
 - Principal-Agent risks
 - Fraud, disclosure, and anonymity risk
 - Smart contract risks
 - Oracle risks

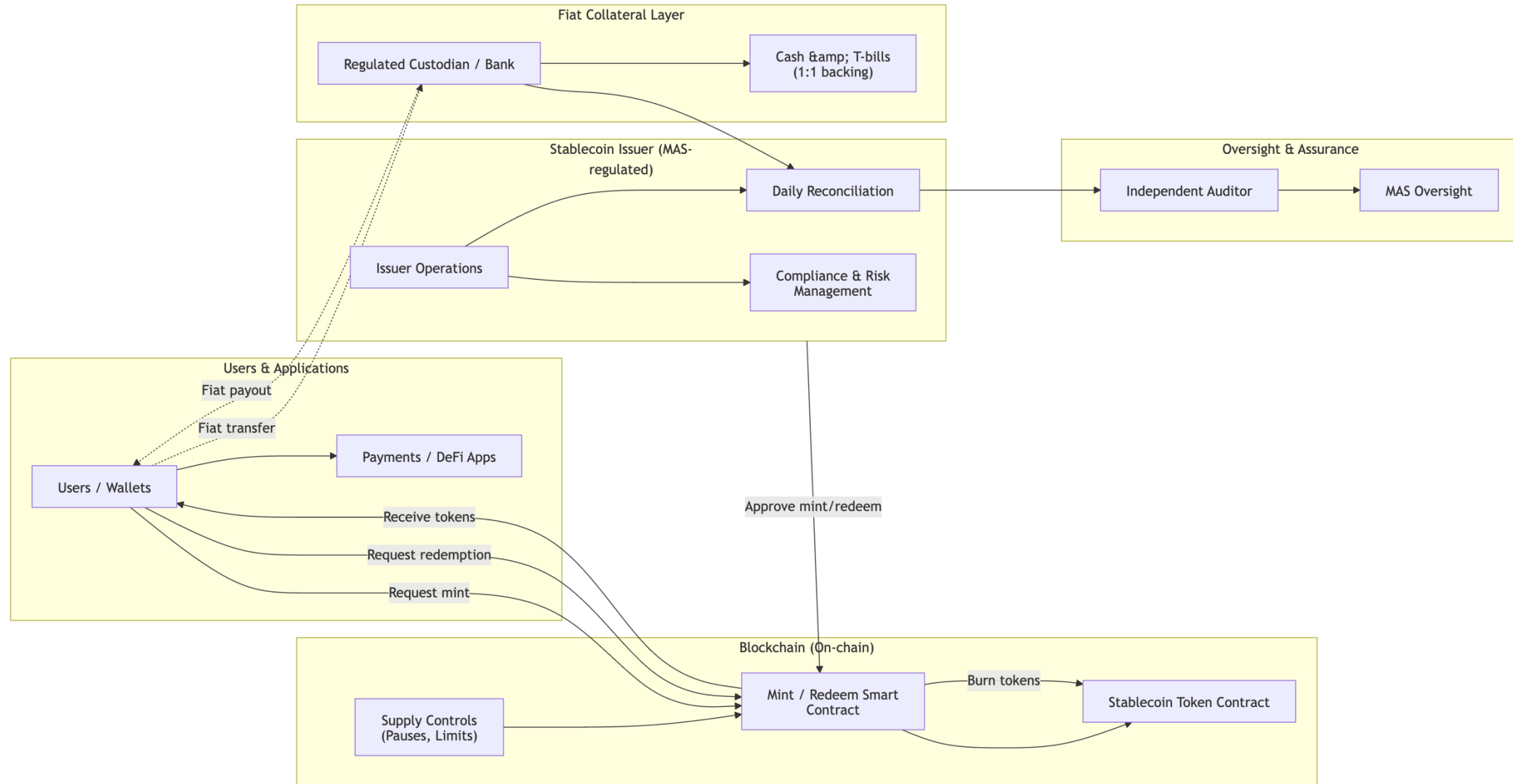
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Stablecoin Design Taxonomy

Category	Core Peg Mechanism	Example
Fiat-Collateralized	Off-chain reserves, issuer redemption	USDC, USDT
Crypto-Collateralized (Algorithmically Governed)	Over-collateralized crypto + on-chain rules	DAI
Pure Algorithmic (Unbacked)	Supply reflexivity & incentives only	UST (failed), Basis
Elastic Supply (Rebase)	Balance adjustment, not price stability	AMPL
Partially Algorithmic / Hybrid	Collateral + algorithmic market ops	FRAX

Fiat-Collatorised Stablecoin



Crypto-Collatorised Stablecoin

Use Case & Problem Definition

- Problem being solved: Need for a price-stable digital currency usable on public blockchains
- Users want:
 - Stability (unit of account, medium of exchange)
 - On-chain composability (DeFi integration)
 - Censorship resistance and self-custody
 - No trusted central issuer
 - Users do not want to rely on banks or off-chain reserves
- Why this problem exists
 - Native crypto assets (ETH, BTC) are volatile.
 - Fiat-backed stablecoins introduce:
 - Custodial risk
 - Censorship and freezing
 - Regulatory dependency



Crypto-Collatorised Stablecoin

Key risks

- Technical
 - Smart contract bugs
 - Oracle failure
- Governance
 - Key compromise
 - Abuse of pause powers
- Market
 - Liquidity fragmentation
 - Trust collapse after incident
- Regulatory
 - Commodity vs security classification
 - Cross-border redemption rules

Crypto-Collatorised Stablecoin

Why blockchain is appropriate

- Stablecoin must be:
 - Globally accessible
 - Verifiable in real time
 - Composable across protocols
- Why decentralisation adds value
 - Collateral and issuance logic are:
 - Transparent
 - Programmatic
 - Enforced by smart contracts
 - No single entity controls minting or redemption

Crypto-Collatorised Stablecoin: Architecture

On-chain Components

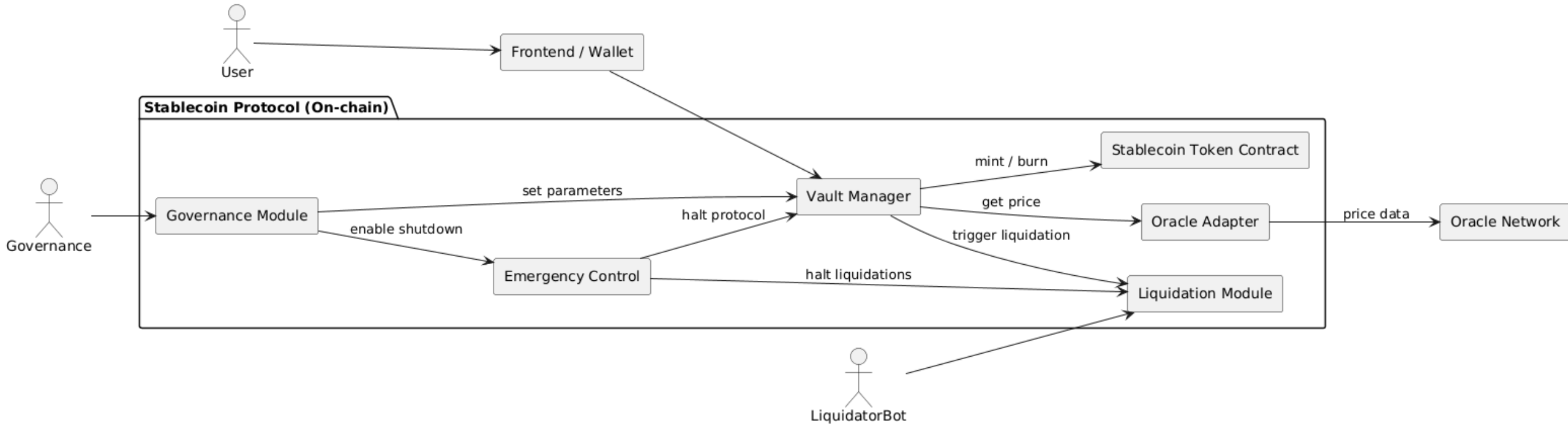
- Vault Manager (Collateral vault contracts)
 - Hold crypto assets (e.g., ETH, stETH)
 - Enforce collateral ratio rules
 - Liquidate when necessary
- Stablecoin smart contracts
 - Mint/burn stablecoins (e.g., DAI)
- Oracle Adaptor Contract (Price feeds integration)
 - Reference oracle values
- Governance contracts
 - Parameter setting (fees, ratios)
- Example:
 - MakerDAO's Vaults + DAI contracts

Crypto-Collatorised Stablecoin: Architecture

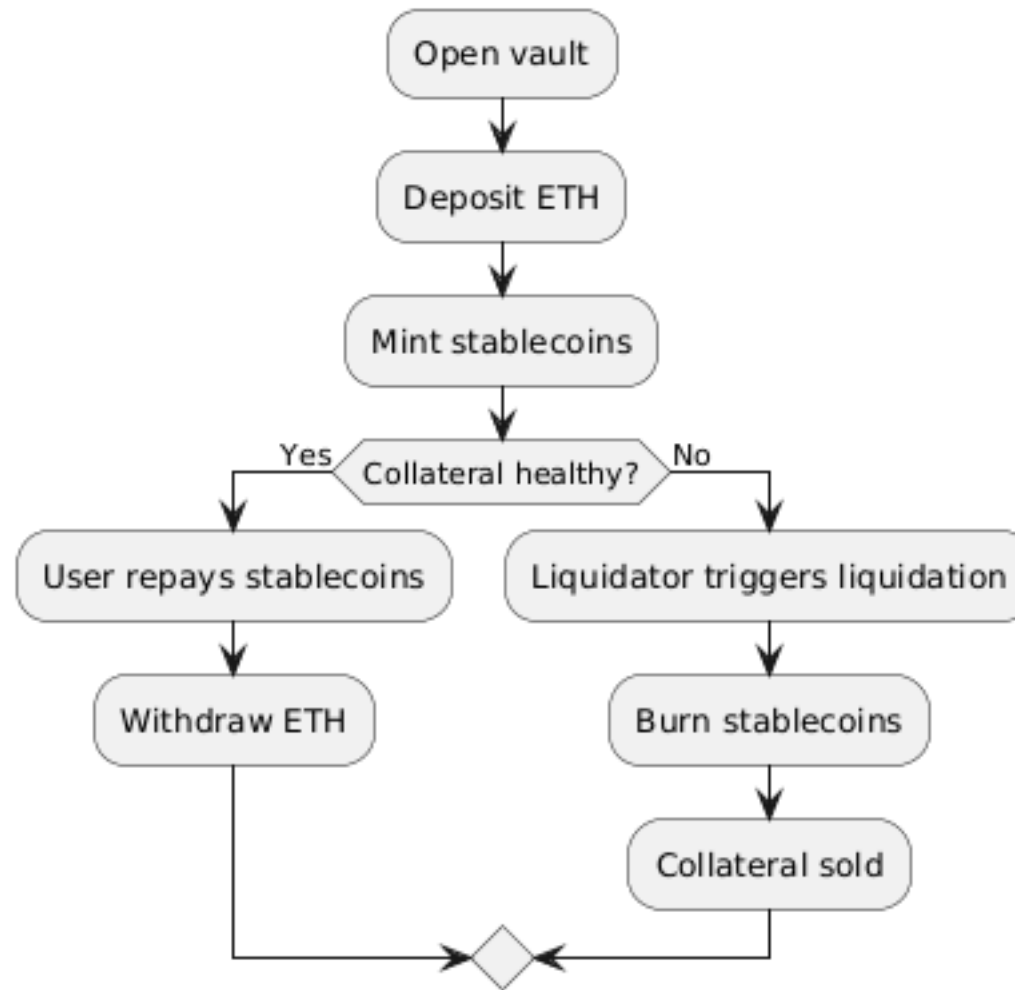
Off-chain Components

- Price oracles
 - Aggregated market prices (e.g., Chainlink)
- User interfaces
 - Web apps, wallets
- Governance coordination
 - Forums, Snapshot votes
- Liquidator(Bot)
 - Risk modelling performed off-chain, enforced on-chain

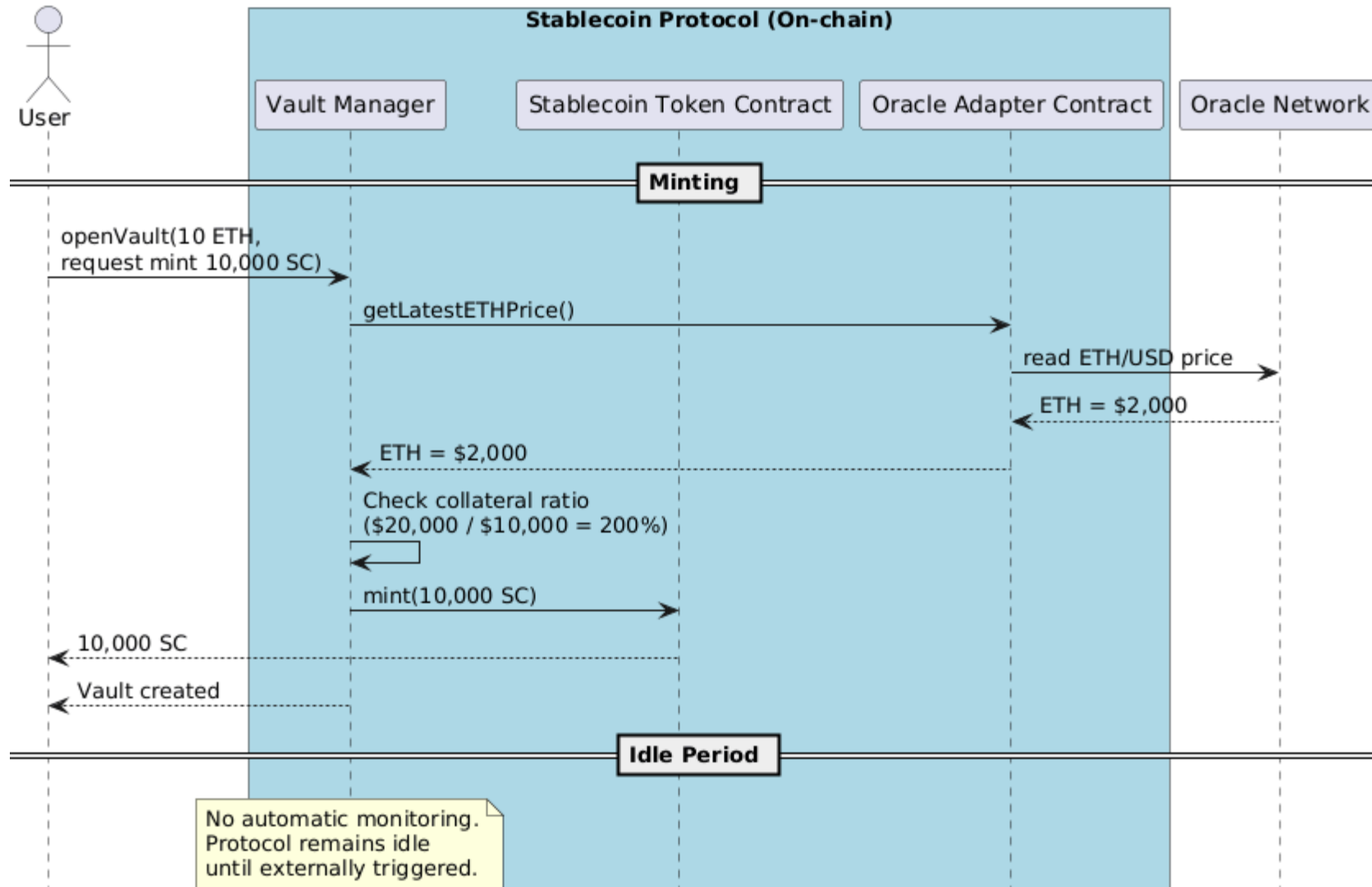
Crypto-Collatorised Stablecoin: Architecture



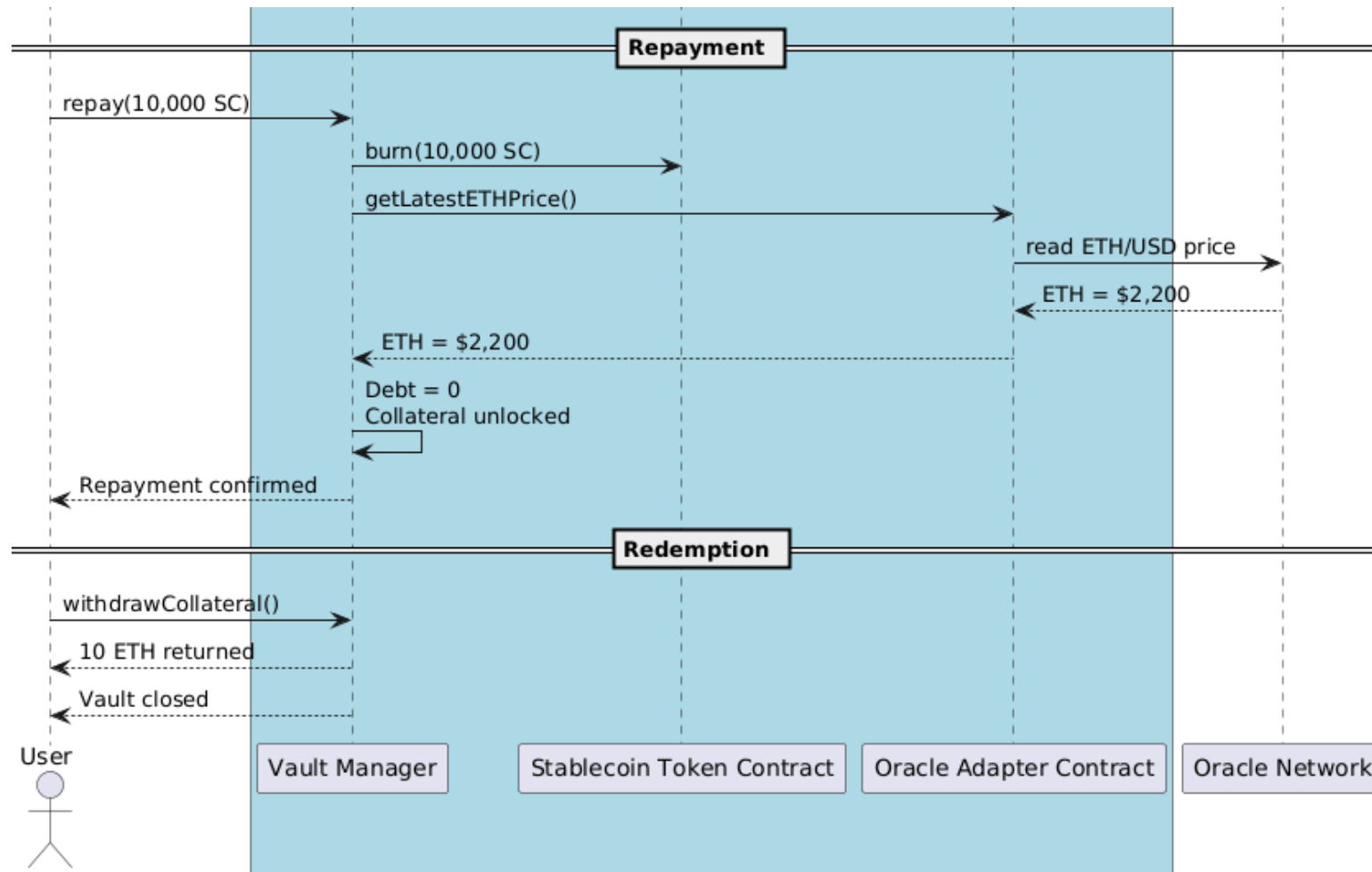
Crypto-Collatorised Stablecoin: Process Flow



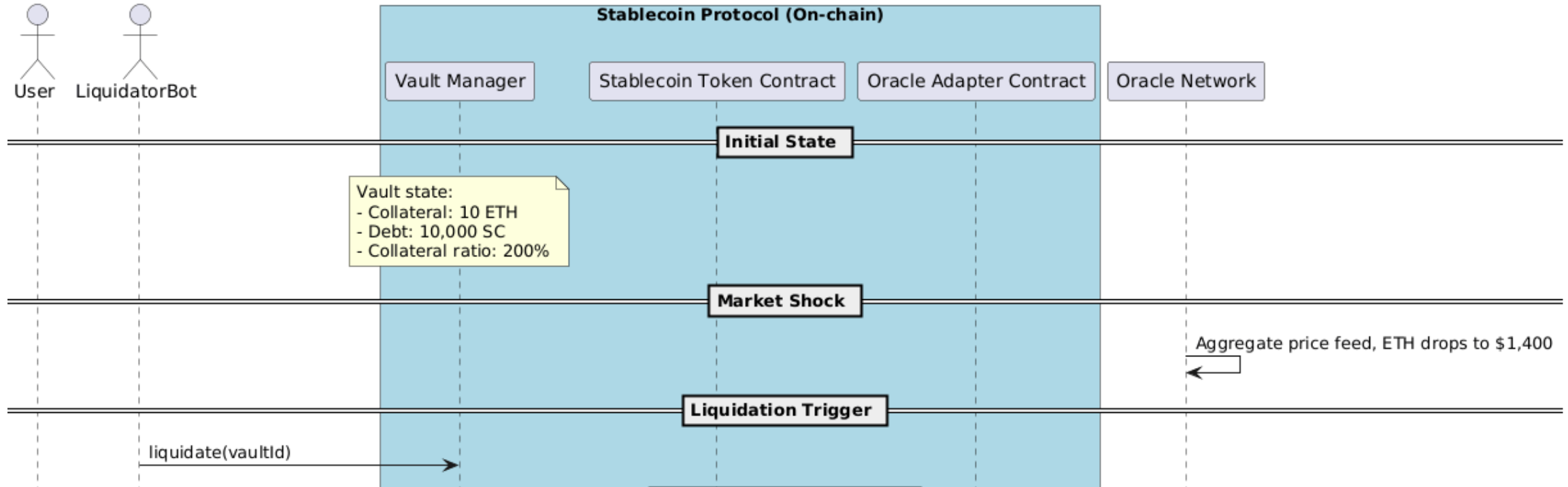
Crypto-Collatorised Stablecoin: Happy Flow



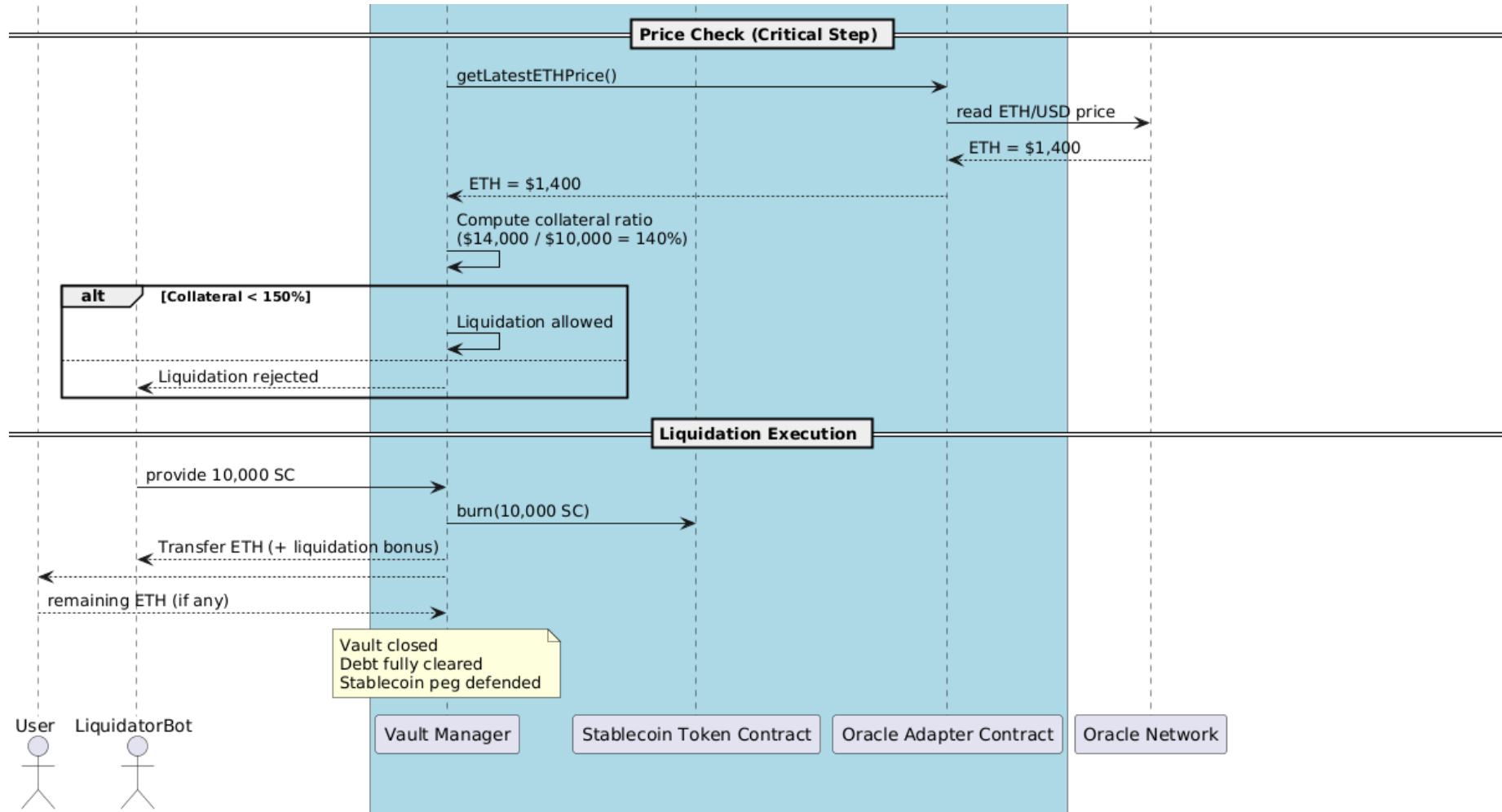
Crypto-Collatorised Stablecoin: Happy Flow



Crypto-Collatorised Stablecoin: Liquidation



Crypto-Collatorised Stablecoin: Liquidation



Comparing with MakerDAO

Dimension	Our Simplified Model	MakerDAO (MCD)
Core Contracts	Single <i>Vault Manager</i> + Stablecoin Token Contract	Multiple contracts (Vat, Urn, Jug, Dog, Clipper, etc.)
Vault Representation	One logical vault per user	Per-collateral vaults (“Urns”)
Collateral Ratio	Single minimum ratio (e.g. 150%)	Collateral-specific ratios set by governance
Price Source	Single oracle adapter (spot price)	Layered oracle system (medianisation + delay via OSM)
Liquidation Method	Direct liquidation (abstracted)	Auction-based liquidation
Fees / Interest	Ignored or assumed flat	Stability fees accrue over time + DAI Savings Rate
Governance	Not modeled	On-chain & off-chain MKR governance
Emergency Controls	Not modeled	Global Settlement / Emergency Shutdown

Crypto-Collatorised Stablecoin

- Incentives
 - Borrowers
 - Access liquidity without selling crypto
 - Liquidators
 - Earn profit from under-collateralised positions
 - Governance token holders
 - Maintain system solvency and credibility
- Economic Design
 - Over-collateralisation (e.g., 110%–170%)
 - Stability fees / borrowing costs
 - Liquidation penalties to discourage risk
- Governance
 - Collateral types
 - Minimum ratios
 - Fees
 - Upgrade mechanisms via on-chain votes

Crypto-Collatorised Stablecoins

- Benefits
 - Non-custodial and censorship-resistant
 - Fully transparent reserves
 - Deep DeFi composability
 - No reliance on banks
- Trade-offs
 - Capital inefficiency (over-collateralisation)
 - Complexity for users
 - Dependence on market liquidity
- Challenges
 - Black-swan market crashes
 - Oracle failures or manipulation
 - Governance capture (for governed systems)
 - Scaling to large supply without centralised assets



Algorithmic Stablecoins

Use Case & Problem Definition

- A USD-denominated digital unit usable on-chain
 - Without reliance on fiat reserves, banks, or custodians
 - While remaining scalable and capital-efficient
 - In short: a decentralized “dollar” without dollars.
- Traditional fiat-backed stablecoins (e.g. USDC, USDT) require:
 - Trust in a centralized issuer
 - Trust in off-chain reserves
 - Trust in regulators and banking rails
- Capital inefficiency of over-collateralization (e.g. DAI)

Algorithmic Stablecoins

Suitability for Blockchain

- Blockchain was appropriate because the design required:
 - Transparent monetary rules
 - Programmatic mint/burn operations
 - Permissionless arbitrage
 - Composable money across DeFi

Algorithmic Stablecoins: Terra

- Terra was a dual-token monetary system:
 - UST → stable unit of account
 - LUNA → volatile absorber and security token
 - There was no collateral pool guaranteeing redemption.

On-Chain Components

- USDT Stablecoin and LUNA Native Token (Mint/Burn Contract)
 - Allow users to mint UST by burning \$1 worth of LUNA
 - Allow users to burn UST to mint \$1 worth of LUNA
 - Enforce deterministic exchange rates via oracle prices
 - Peg maintenance logic:
 - If $UST > \$1$
 - Mint UST by burning \$1 of LUNA
 - Increases UST supply → price falls toward \$1
 - If $UST < \$1$
 - Burn UST to mint \$1 of LUNA
 - Reduces UST supply → price rises toward \$1
 - Traders profit from price deviations
 - Expected to keep the peg stable

Algorithmic Stablecoins: Terra

On-Chain Components

- Oracle Contracts
 - Provided USD prices for LUNA
 - LUNA is traded on both centralized and decentralized exchanges
- Staking & Governance
 - LUNA stakers secured the chain
 - Governance controlled parameters (fees, thresholds)
- Anchor Protocol: Major demand sink for UST

Off-Chain Components

- Exchanges / DEXs: Provided price discovery and liquidity
- Arbitrage bots: Enforced the peg incentive
- Market makers: Provided off-chain LUNA ↔ USD liquidity
- LUNA Foundation Guard

Algorithmic Stablecoins: Terra

- Anchor Protocol was a lending and borrowing DeFi application on the Terra blockchain, designed to make UST attractive as a yield-bearing stablecoin.
 - Create persistent demand for UST
 - Offer a stable, predictable yield (~19–20% APY) on UST deposits
 - Position itself as a “crypto savings account”
- In practice, Anchor became:
 - The main sink of UST liquidity
 - The center of gravity of the entire Terra ecosystem
- Core Participants in Anchor
 - UST Depositors (Lenders)
 - ~19–20% APY paid in UST
 - Borrowers
 - Borrowed UST from Anchor
 - Using bLUNA (bonded LUNA)

Algorithmic Stablecoins: Terra

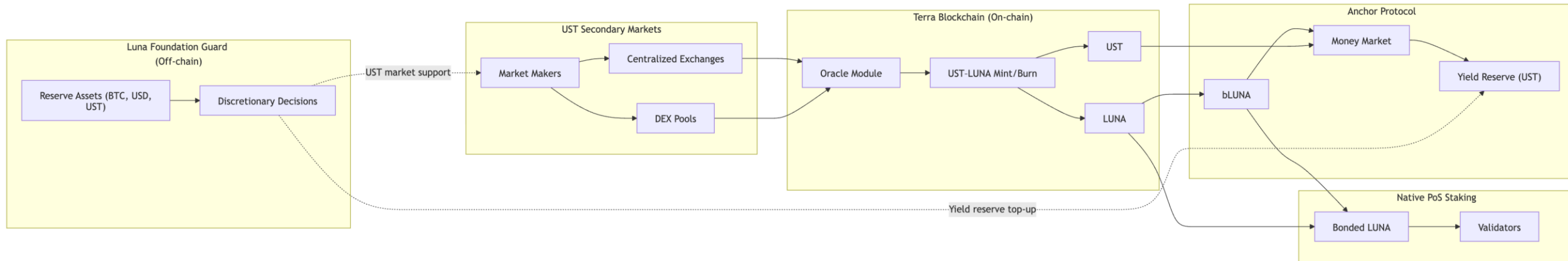
- LUNA was a Proof-of-Stake (PoS) token
- Staking LUNA:
 - Secured the Terra blockchain
 - Generated staking rewards
- Borrowers bonded their LUNA
 - Bonded LUNA = bLUNA
 - Represented staked LUNA
 - Continued earning staking rewards
 - Could be used as collateral

Algorithmic Stablecoins: Terra

- Luna Foundation Guard (LFG) was a nonprofit set up to:
 - Strengthen the UST peg
 - Act as a last-resort defender
- Accumulated reserves, mainly:
 - Bitcoin (BTC)
- Intended function:
 - Sell reserves to buy UST during de-pegging events
 - LFG (and Terraform Labs) replenished Anchor's Yield Reserve
 - These injections were discretionary and not protocol-enforced
 - Were explicitly publicized to maintain confidence in Anchor

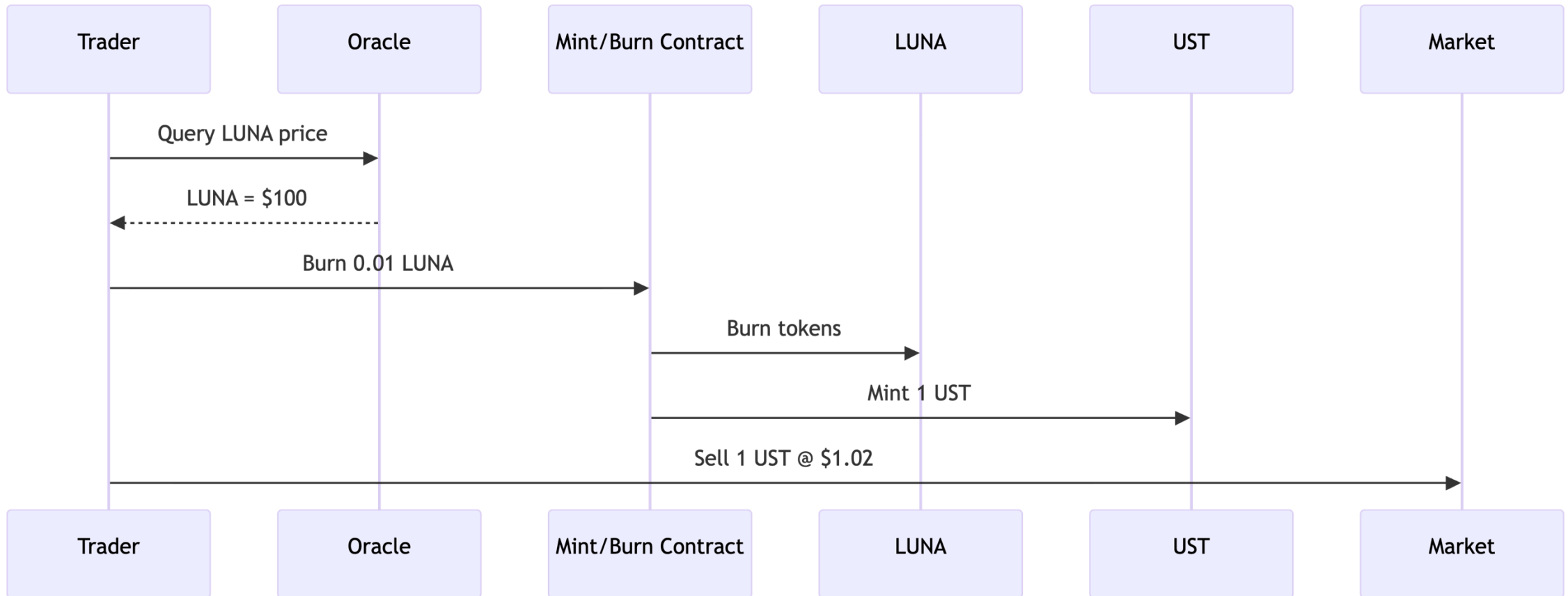
Algorithmic Stablecoins: Terra

System Diagram



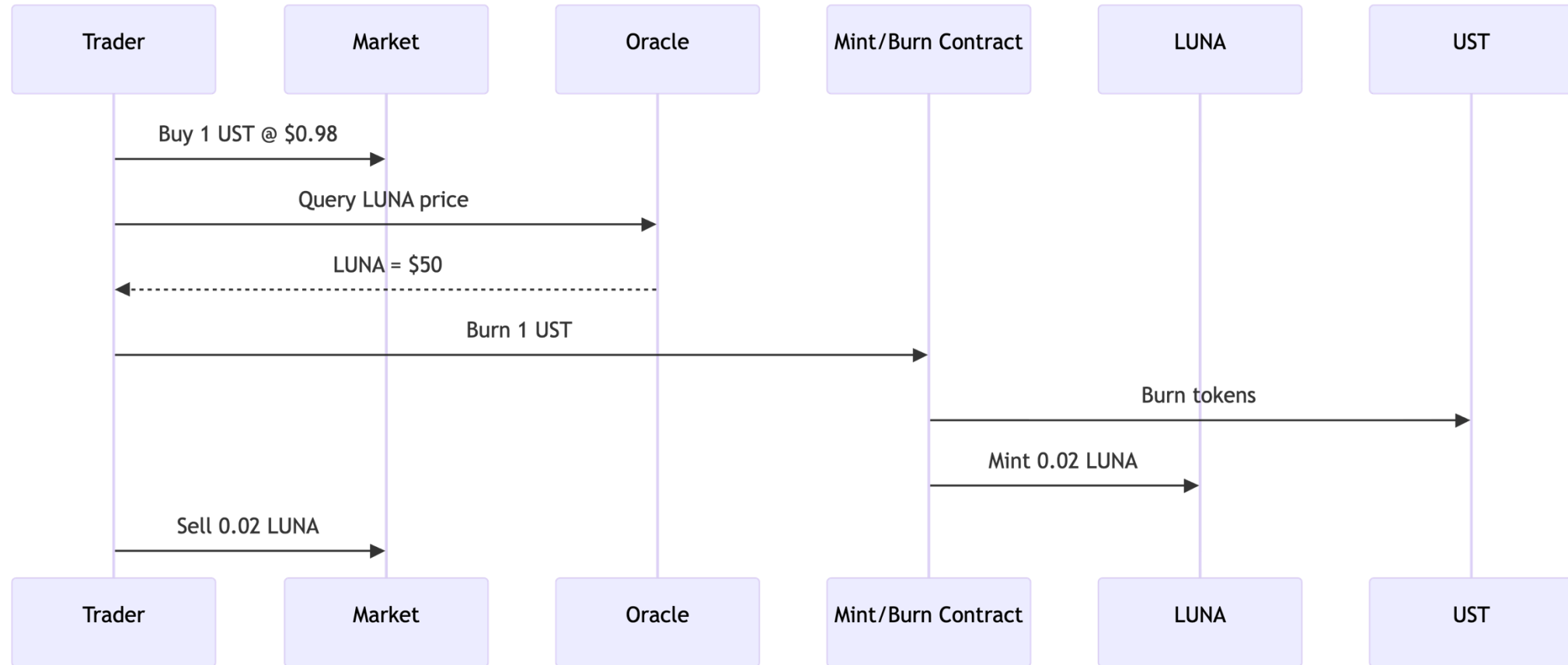
Algorithmic Stablecoins: Terra

Burning LUNA for UST



Algorithmic Stablecoins: Terra

- Burning UST for LUNA



Algorithmic Stablecoins: Terra

Death Spiral Scenario

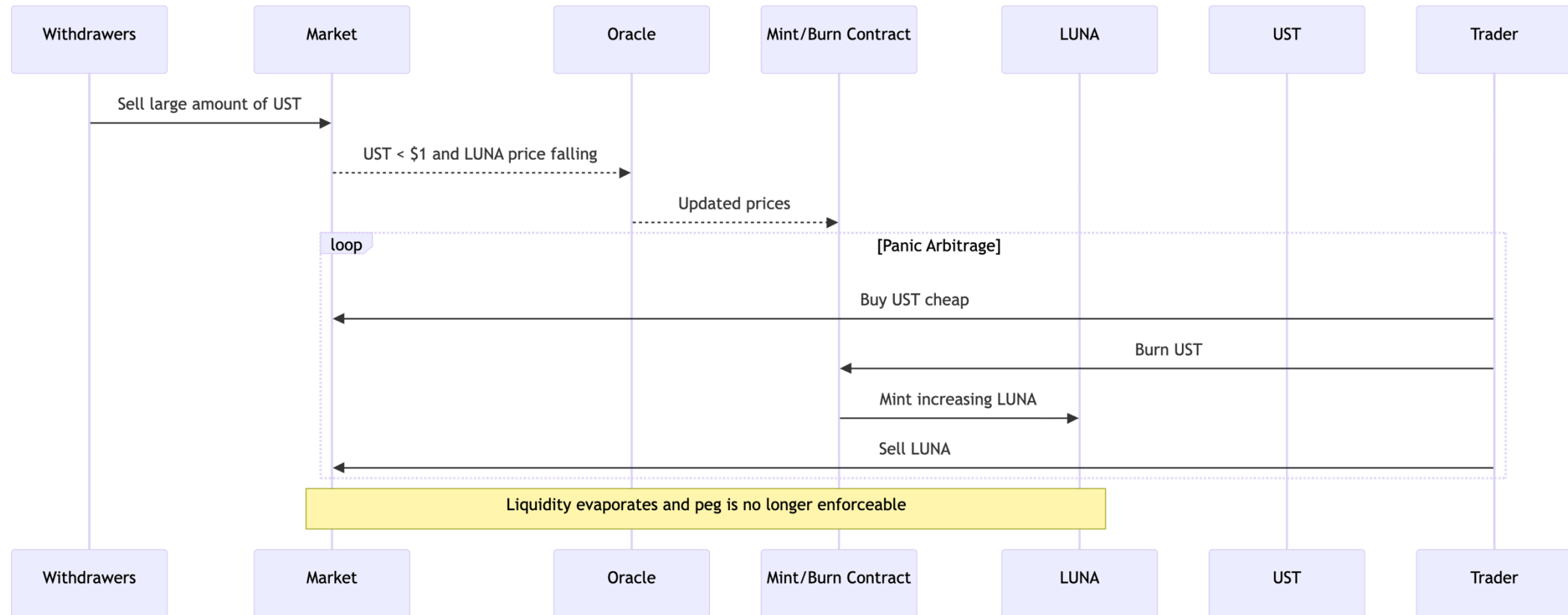
- Large UST exits (Anchor withdrawals)
- Massive UST → LUNA conversions
- LUNA supply inflates quickly
- LUNA price collapses
- Minted LUNA $\neq$ \$1 anymore
- Arbitrage breaks

- Numerical illustration:
 - LUNA price \$80 → \$20 → \$5
 - To redeem \$1 UST:
 - Mint 0.0125 → 0.05 → 0.20 LUNA

- Supply explodes
- Confidence collapses
- Peg no longer enforceable
- There is no terminal floor

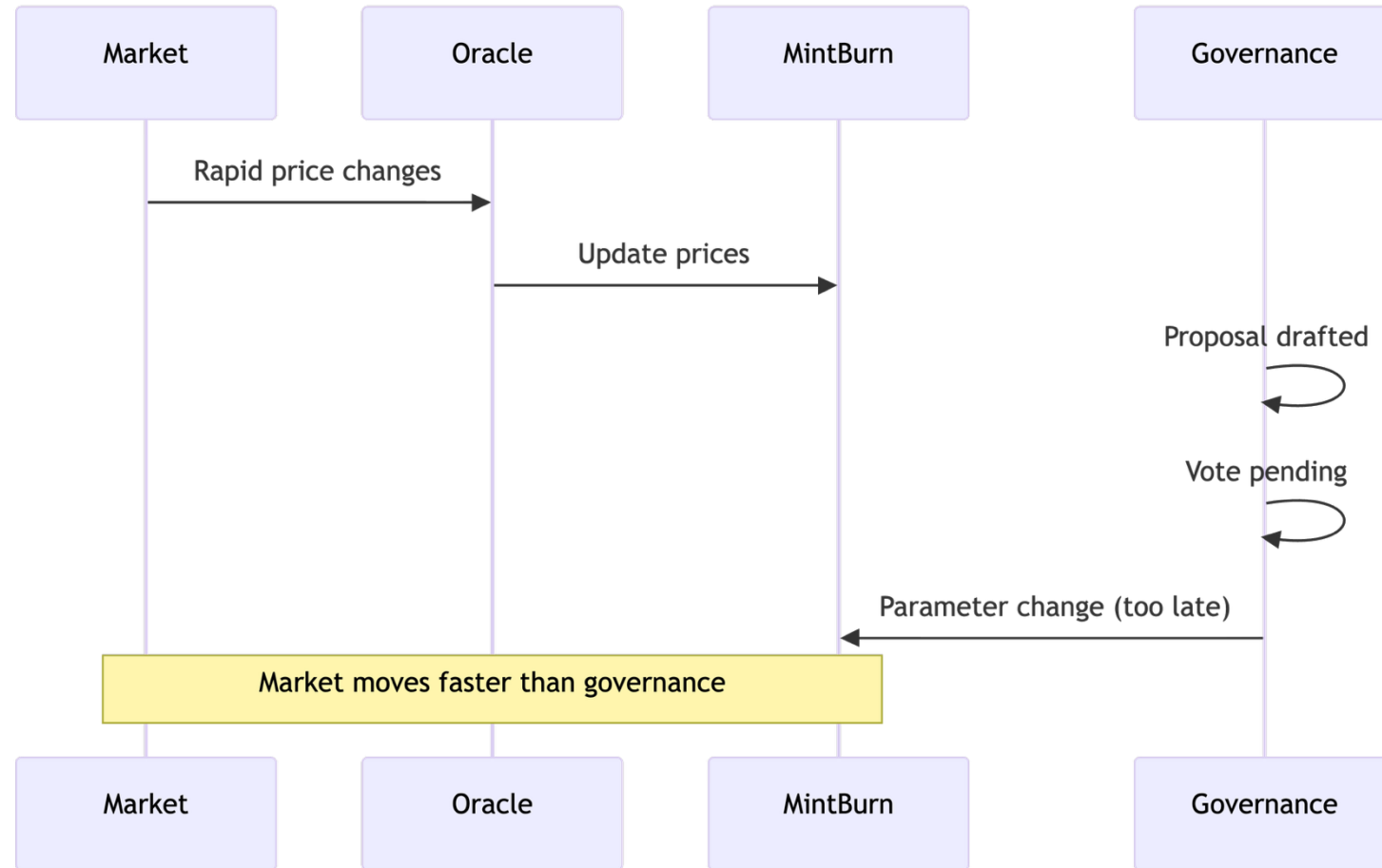
Algorithmic Stablecoins: Terra

Death Spiral



Algorithmic Stablecoins: Terra

Death Spiral: Governance (LFG)



Algorithmic Stablecoins: UST Depegging

- UST Decoupling: On May 10, 2022, UST started decoupling from its dollar peg
- Value Plunge: UST dropped to below \$0.30 on May 11, losing a third of its value
- Causes of the Crash:
 - Luna Foundation Guard (LFG): LFG raised \$1 billion for a BTC reserve for UST, leading to high-basis UST minting
 - This triggered a massive run-up in LUNA's price, allowing for high-basis UST minting
 - LUNA hit an all-time high of \$119 in early April 2022
 - High-basis UST meant its value was more than \$1
 - People started selling UST for profit, causing its value to drop
 - UST decoupled from its \$1 peg, like a train coming off its tracks
 - LUNA Price Volatility: LUNA's price surged and then crashed, affecting UST stability

Algorithmic Stablecoins: Terra

- Incentives
 - Arbitrageurs: profit from peg deviations
 - LUNA holders: benefit from UST growth (supply burn)
 - Validators: earn fees, secure chain
- Failure of Incentives
 - In contraction, arbitrage creates sell pressure
 - LUNA holders become dilution victims
 - Rational actors exit instead of stabilizing
- Governance & Controls
 - Governance could adjust fees, limits
 - No automatic shutdown or circuit breaker
 - No capped issuance under stress
 - Ultimate power:
 - Protocol governance + social consensus
 - But too slow vs market speed.

Algorithmic Stablecoins: Terra

- Benefits
 - Capital-efficient (no idle collateral)
 - Fully on-chain enforcement
 - Permissionless mint/burn
 - No custodians or banks
- Trade-offs
 - Stability depends entirely on confidence
 - Reflexive feedback loops
 - No balance sheet
 - Incentives reverse under stress
- Risks
 - Market Risk
 - Liquidity evaporation
 - Correlated exits
 - Technical Risk
 - Oracle lag or failure
 - MEV extraction during arbitrage windows
 - Governance Risk
 - Delayed response
 - Politicized interventions
 - Structural Risk (Core)
 - Unbounded supply expansion without a redemption floor

Group Discussion (Part of Participation marks)

- Terra's failure revealed that its stability depended on confidence, liquidity, and off-chain discretion rather than hard constraints
- Your task: Redesign the Terra system so that a Terra-like stablecoin could plausibly survive a large negative demand shock without requiring discretionary bailouts or a death spiral
- This should be based on Terra's original design and not copying DAI or USDT
- What to Present (≤ 5 minutes)
 - Your redesign in one sentence
 - One diagram or short flow description
 - The main trade-off you chose and why

Comparative Analysis using Simulations

- The paper adopt an agent-based modeling approach and simulate each category
- By varying different parameters and conditions in the simulations, they aim to identify under what circumstances each type of stablecoin is stable and not stable
- They adopt a categorization of stablecoins as shown below

Collateral source	Collateral management	
	Central	Decentral
Exogenous	USDT, ²⁰ USDC ²¹	Dai ²²
Endogenous	TerraUSD ²³	sUSD ²⁴

Table 1. 2×2 matrix for categorizing stablecoins with examples of each type.

Hafner, M., Pereira, M. H., Dietl, H., & Beccuti, J. (2023). The four types of stablecoins: A comparative analysis. *arXiv preprint arXiv:2308.07041*.

Comparative Analysis using Simulations

- The study highlights potential risks associated with stablecoins with endogenous and centrally managed collateral, as they may be at a higher risk of crashing after a demand shock
 - A sudden surge in demand can trigger a death spiral, as seen in the TerraUSD crash
 - It is also shown that central collateral management may pose a greater risk of a bank run than decentralized collateral management because users may withdraw their funds en masse, leading to a collapse in the stablecoins' value
 - Therefore, on one hand, stablecoins with exogenous and decentrally managed collateral, like Dai, may be considered the safest choice

Hafner, M., Pereira, M. H., Dietl, H., & Beccuti, J. (2023). The four types of stablecoins: A comparative analysis. *arXiv preprint arXiv:2308.07041*.

Comparative Analysis using Simulations

- On the other hand, stablecoins with endogenous and decentrally managed collateral, such as sUSD, provide greater autonomy and independence
- However, these two designs also come at a cost: without additional measures, Dai and sUSD are more expensive to create and experience stronger fluctuations around their peg than others
- The authors conclude that the choice between different types of stablecoins depends on individual risk preferences and specific use cases and that a trilemma exists between stability, independence, and costs

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Comparative Analysis using Simulations

- The study emphasizes the importance of carefully evaluating the source of a stablecoin's collateral and its collateral management mechanism to ensure stability and minimize risks.
- The authors recommend that policymakers should exercise caution when dealing with stablecoins, such as TerraUSD, that have endogenous and centrally managed collateral
- To protect users and mitigate systemic risks in the financial system, it is crucial for policymakers to take steps to ensure the stability of these stablecoins through appropriate regulations
- However, stablecoins that use a combination of endogenous and decentrally managed collateral, such as sUSD, have proven stable and reliable
- Therefore, regulators should carefully consider both dimensions when crafting policies rather than imposing overly restrictive measures on all stablecoins with endogenous collatera.

Hafner, M., Pereira, M. H., Dietl, H., & Beccuti, J. (2023). The four types of stablecoins: A comparative analysis. *arXiv preprint arXiv:2308.07041*.



Wooclap Quiz

